

Lesson 16

Design Goals

- **Edward Tufte's Data-Ink Ratio Principle:** Maximize the data-ink ratio, meaning every pixel of ink on a graphic should present new information. Non-data pixels (such as background or decoration) should be minimized.
- **Reduce Non-Data Pixels:** Eliminate unnecessary elements and de-emphasize the remaining ones to focus on the data.
- **Enhance Data Pixels:** Remove unnecessary data elements and highlight the most important data for clarity.

Example of Reducing Non-Value Adding Elements:

- 3D diagrams, grid lines in bar graphs, unnecessary decoration, color gradients, and irrelevant background.
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Visual Design Process

- **Top-Left and Center Areas:** These areas receive the most visual attention due to natural reading patterns and the way human vision works.
 - **Emphasize critical information** in these prime areas.
 - Avoid placing non-essential items (e.g., logos) in these areas.
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Use of Visual Attributes to Enhance Designs

1. **Color Intensity:** Saturated colors are perceived as more important.
 2. **Size:** Larger elements are perceived as more important.
 3. **Line Width:** Thicker lines stand out more and signify importance.
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Design Goals for Dashboards

- **Simple Presentation:** Display a large amount of data in a small space without clutter.
- **Effectiveness and Expressiveness:** The design should clearly communicate information.
- **Space Efficiency:** Dashboards should still be effective even when scaled to smaller spaces.

Example: OECD Life Expectancy Dashboard: [OECD Gender Dashboard](#)

Categorization of Design Visuals

- **Graphs:** Line, bar, area graphs.
- **Images:** Photographs and illustrations.
- **Icons:** Represent on/off, right/wrong, increase/decrease.
- **Text:** Highlight key elements.
- **Organizers:** Small multiples for comparative views.
- **Tables:** Used for precise data presentation.
- **Spatial Maps:** To show geographic relationships.

Example: [OECD Life Expectancy Dashboard](#)
